RACE Regular Analysis Through Continuous Exercise

Coordination Compounds : RACE-01

Types of Ligands: On the basis of Charge

1. Which one of the following is NOT a ligand ?

(A) PH_3 (B) $NO^{\oplus}$ (C) $Na^{\oplus}$ (D) $F^{\ominus}$

CCC123

Types of Ligands: On the basis of Denticity; Ligand as Polydentate Ligand; Chelating Ligands; Ambidentate Ligands; Flexidentate Ligands;

2. en (ethylene diamine) is a

(A) Monodentate ligand (B) Bidenate ligand

(C) Tridentate ligand (D) Hexadentate ligand

CCC126

3. The donor sites of $(EDTA)^{4-}$ are ?

(A) O atoms only (B) N atoms only

(C) Two N atoms and four O atoms (D) Three N atoms and three O atoms

CCC354

4. How many of the following ligands are monodentate ligands

(A) CH_3NC (B) $acac^{-1}$ (C) N_3^- (D) dien

(E) $edta^{4-}$ (F) $edta^{3-}$ (G) ox^{2-} (H) dmg^{-1}

(I) NC^- (J) NO_2^- (K) O^{2-} (L) O_2^-

CCC355

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Werner Theory; Electrical and Molar Conductivity, Colligative Properties of aq. Solution of Coordination Compounds

5. Which of the following will exhibit maximum ionic conductivity?

(A) $K_4[Fe(CN_6]$ (B) $[Co(NH_3)_6]Cl_3$ (C) $[Cu(NH_3)_4]Cl_2$ (D) $[Ni(CO)_4]$

CCC356

Effective Atomic Number; Calculation of EAN; Sidgwick's EAN Rule; Effective Atomic Number of Bridging Complexes; Oxidising and Reducing Property of Carbonyl Complex

6. Which of the following complex is following sidwick EAN rule.

(A) $[Ag(S_2O_3)_2]^{3\ominus}$ (when only 'S' atom is the donor atom)

(B) $[Cd(CN)_4]^{2\ominus}$

(C) $[Pt(en)_2]^{2\oplus}$

(D) $[Mo(\sigma- C_3H_5)Br(NH_3)_2]^0$

CCC164

7. Determine the effective atomic number of the metal atom in $K_4[Fe(CN)_6]$

CCC169

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8. Oxidation number of Fe in violet coloured complex $Na_4[Fe(CN)_5(NOS)]$ is :

(A) 0 (B) 2 (C) 3 (D) 4

CCC357

9. The oxidation number of Co in the complex ion $[(en)_2Co\begin{matrix}NH\\OH\end{matrix}Co(en)_2]^{3+}$ is :

(A) +2 (B) +3 (C) +4 (D) +6

CCC358

10. The EAN of metal atoms in $[Fe(CO)_2(NO^+)_2]$ and $Co_2(CO)_8$ respectively are :

(A) 34, 35 (B) 34, 36 (C) 36, 36 (D) 36, 35

CCC359

11. The effective atomic number of $Co(CO)_4$ is 35 and hence is less stable. It attains stability by

(A) Oxidation of Co (B) Reduction of Co (C) Dimerization (D) None

CCC360

12. In the complex $[CoCl_2(en)_2]$ Br, the co-ordination number and oxidation state of cobalt are :

(A) 6 and +3 (B) 3 and +3 (C) 4 and +2 (D) 6 and +1

CCC361

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	C	6	A	B	36	B	B	C
Question	11	12								
Answer	BC	A								

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RACE Regular Analysis Through Continuous Exercise

Coordination Compounds : RACE-02

IUPAC Nomenclature of Coordination Compounds

1. The formula of the complex tris(ethylenediamine)cobalt(III) sulphate is :

(A) $[Co(en)_2SO_4]$ (B) $[Co(en)_3SO_4]$

(C) $[Co(en)_3]_2SO_4$ (D) $[Co(en)_3]_2(SO_4)_3$

CCC362

2. The correct IUPAC name for the compound $[Co(NH_3)_4Cl(ONO)]Cl$ is :

(A) Tetraamminechloridonitrito–N–cobalt(III) chloride

(B) Chloridonitrito–O–tetraamminecobalt(II) chloride

(C) Dichloridonitrito–O–tetraamminecobalt(III)

(D) Tetraamminechloridonitrito–O–cobalt(III) chloride

CCC363

3. The hypothetical complex triamminediaquachloridocobalt(III) chloride can be represented as :

(A) $[CoCl(NH_3)_3(H_2O)_2]$ (B) $[Co(NH_3)_3(H_2O)Cl_3]$

(C) $[Co(NH_3)_3(H_2O)_2Cl]Cl_2$ (D) $[Co(NH_3)_3(H_2O)_3]Cl_3$

CCC364

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4. How many total sodium ions are present in one formula unit of sodium ethane-1,2-diaminetetraacetatochromate(II) and sodium hexanitritocobaltate(III) ?

CCC365

Structural Isomerism

5. In Coordination compounds, the hydrate isomers differ :-

(A) In the number of water molecules of hydration only

(B) In the number of water molecules only present as ligands

(C) Both in (A) and (B)

(D) In their Coordination number of the metal atom

CCC194

6. Which kind of isomerism is shown by the complex $[Co(NH_3)_5(ONO)]SO_4$?

1. Ionisation isomerism 2. Linkage isomerism

3. Geometrical isomerism 4. Optical isomerism

(A) 1, 2, 3 and 4 are correct (B) 1, 3 and 4 are correct only

(C) 1 and 2 are correct only (D) 2, 3 and 4 are correct only

CCC366

7. The complexes $[Pt(NH_3)_4]$ $[PtCl_6]$ and $[Pt(NH_3)_4Cl_2]$ $[PtCl_4]$ are :

(A) Linkage isomers (B) Optical isomers

(C) Co-ordination isomers (D) Ionisation isomers

CCC367

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Stereoisomerism for Square Planar Complexes

8. Cis-trans isomerism is found in square planar complexes of molecular formula ('a' and 'b' are monodentate ligands) :

(A) Ma_4 (B) Ma_3b (C) Ma_2b_2 (D) Mab_3

CCC368

Stereoisomerism for Octahedral Complexes for Monodentate Ligand

9. The number of geometrical isomer of $[Co(NH_3)_3(NO_2)_3]$ are :

(A) 0 (B) 2 (C) 3 (D) 4

CCC369

10. Geometrical isomerism is found in coordination compounds having coordination number :

(A) 2 (B) 3 (C) 4 (tetrahedral) (D) 6

CCC370

11. The number of geometrical isomers for octahedral $[Co(NH_3)_2Cl_4]^-$, square planar $AuCl_2Br_2^-$ are :

(A) 4, 2 (B) 2, 2 (C) 3, 2 (D) 2, 3

CCC371

Stereoisomerism for Octahedral Complexes for Symmetrical Bidentate Ligand

12. Which of the following complex will show optical activity ?

(A) trans-$[Co(NH_3)_4Cl_2]^+$ (B) $[Cr(H_2O)_6]^{3+}$

(C) cis-$[Co(NH_3)_2(en)_2]^{3+}$ (D) trans-$[Co(NH_3)_2(en)_2]^{3+}$

CCC372

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	D	C	5	C	C	C	C	B	D
Question	11	12								
Answer	B	C								

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RACE Regular Analysis Through Continuous Exercise

Coordination Compounds : RACE-03

Bonding in Coordination Compound; Valence Bond Theory; Limitation of Valence Bond Theory

1. In the co-ordination compound $Na_4[Fe(CN)_5NOS]$ oxidation state of Fe is -

(A) +1 (B) +2

(C) +3 (D) +4

CCC253

Spectro chemical series (General order of strength of ligands)

2. The crystal field-splitting for Cr^{3+} ion in octahedral field changes for ligands I^-, H_2O, NH_3, CN^- and the increasing order is :

(A) $I^- < H_2O < NH_3 < CN^-$ (B) $CN^- < I^- < H_2O < NH_3$

(C) $CN^- < NH_3 < H_2O < I^-$ (D) $NH_3 < H_2O < I^- < CN^-$

CCC373

Crystal Field Theory for Octahedral Complexes, Hybridisation and Geometry of Octahedral Complexes

3. The CORRECT statement for complex ammine of Fe(II) and Fe(III) is

(A) Both Fe(II) and Fe(III) form outer orbital complex with NH_3

(B) Both form inner orbital complex with NH_3

(C) Fe(II) form outer while Fe(III) form inner orbital complex with NH_3

(D) All are incorrect statements regarding the complex

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CCC229

4. According to C.F.T., ligands are treated as -

(A) Point charges (B) Lewis acids

(C) Proton donor (D) All of the above

CCC265

5. Select appropriate ligand for given complex

$\left[\underset{\text{Co}(.....)_6}{\overset{\text{III}}{}}\right]^{x\oplus}$; $\mu = 0$ BM

(A) $C_2O_4^{2\ominus}$ (B) en

(C) H_2O (D) $F\ominus$

CCC270

6. A complex of certain metal has the magnetic moment of 4.91 BM whereas another complex of the same metal with same oxidation state has zero magnetic moment. The metal ion could be :

(A) $Co^{2\oplus}$ (B) $Mn^{2\oplus}$

(C) $Fe^{2\oplus}$ (D) $Fe^{3\oplus}$

CCC374

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Crystal Field Theory for Tetrahedral Complexes, Hybridisation and Geometry of Tetrahedral complexes

7. Out of $[Fe(CN)_6]^{4\ominus}$, $[Ni(CN)_4]^{2\ominus}$ and $[Ni(CO)_4]$

(A) All have identical geometry.

(B) All are paramagnetic.

(C) All are diamagnetic.

(D) $[Fe(CN)_6]^{4\ominus}$ is diamagnetic but $[Ni(CN)_4]^{2\ominus}$ and $[Ni(CO)_4]$ are paramagnetic

CCC254

8. In crystal field theory for complex $[NiCl_4]^{2\ominus}$, the lobes of how many d–orbitals of $Ni^{2\oplus}$ coincide with direction of the ligands ?

(A) zero (B) Two

(C) Four (D) One

CCC275

9. How many of the following is correctly matched complex ?

	Complex	Oxidation no. on central metal	Electronic configuration
(a)	$K_3[Co(C_2O_4)_3]$	+3	t_{2g}^6
(b)	$(NH_4)_2[CoF_4]$	+2	$t_{2g}^5 \; e_g^2$
(c)	Cis - $[Cr(en)_2Cl_2]Cl$	+3	$t_{2g}^3 \; e_g^0$
(d)	$[Mn(H_2O)_6]SO_4$	+2	$t_{2g}^3 \; e_g^2$

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CCC375

Calculation of Crystal Field Stabilisation Energy for Octahedral Complexes

10. Which one of the following high-spin complexes has the largest CFSE (Crystal field stabilization energy)?

(A) $[Mn(H_2O)_6]^{2+}$ (B) $[Cr(H_2O)_6]^{2+}$

(C) $[Mn(H_2O)_6]^{3+}$ (D) $[Cr(H_2O)_6]^{3+}$

CCC376

Crystal Field Theory for Square Planar Complexes, Hybridisation and Geometry of Square planar complexes

11. Correct splitting energy diagram for square planar complex $[PtCl_4]^{2\ominus}$ is

(A) $d_{x^2-y^2}$ ↿⇂ d_{xy} ↿⇂ ↿⇂ d_{yz} d_{xz} ↿⇂ d_{z^2}

(B) $d_{x^2-y^2}$ ↿⇂ d_{xy} ↿⇂ d_{z^2} ↿⇂ ↿⇂ d_{yz} d_{xz}

(C) d_{xz} d_{yz} d_{zx} ↿⇂ 1 1 ↿⇂ ↿⇂ d_{z^2} $d_{x^2-y^2}$

(D) d_{z^2} $d_{x^2-y^2}$ 1 1 ↿⇂ ↿⇂ ↿⇂ d_{xy} d_{yz} d_{zx}

CCC278

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12. Nickel (Z = 28) combines with a uninegative monodentate ligand to form a diamagnetic complex $[NiL_4]^{2\ominus}$. The hybridisation involved and the number of unpaired electrons present in the complex are respectively :

(A) sp^3, zero (B) sp^3, two (C) dsp^2, one (D) dsp^2, zero

CCC282

13. Match List-I (Complexes) with List-II (Hybridization) of central atom and select the correct answer using the codes given below the lists :

	List-I		**List-II**
A	$Ni(CO)_4$	1.	sp^3
B	$[Ni(CN)_4]^{2-}$	2.	dsp^2
C	$[Fe(CN)_6]^{4-}$	3.	sp^3d^2
D	$[MnF_6]^{4-}$	4.	d^2sp^3

	A	B	C	D
(A)	1	3	2	4
(B)	5	2	4	3
(C)	5	3	2	4
(D)	1	2	4	3

CCC377

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ANSWER KEY

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Answer	B	A	C	A	C	C	C	A	3	D
Question	**11**	**12**	**13**							
Answer	B	D	D							

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RACE
Regular Analysis Through Continuous Exercise

Coordination Compounds : RACE-04

Factors Affecting Crystal Field Stabilisation Energy

1. Which of the following statements related to crystal field splitting in octahedralCoordination entities is incorrect ?

(A) The $d_{x^2-y^2}$ and d_{z^2} orbitals has more energy as compared to d_{xy}, d_{yz} and d_{xz} orbitals.

(B) Crystal field spitting energy (Δ_0) depends directly on the charge of the metal ion and on the field produced by the ligand.

(C) In the presence of $Br^{\ominus}$ as a ligand the distribution of electrons for d^4 configuration will be t_{2g}^3, e_g^1

(D) In the presence of $CN^{\ominus}$ as a ligand $\Delta_0 < P$.

CCC286

2. For which of the following ligands magnitude of the crystal field splitting (Δ_0) will be greater than pairing energy (P) ?

(A) $Cl^{\ominus}$ (B) $SCN^{\ominus}$ (C) CO (D) $S^{2\ominus}$

CCC287

Colour in Transition MetalCoordination Compound Using d - d Transition; Charge Transfer Spectrum; Complementary Colour Wheel

3. In given options complex co-ordination entities are written with wavelength of light absorbed for d-d transition.

Select option in which co-ordination entity does not absorb visible light.

(A) $[Co(CN)_6]^{3-}$; 310 nm (B) $[Ti(H_2O)_6]^{3+}$; 498 nm

(C) $[Co(NH_3)_6]^{3+}$; 475 nm (D) $[Co(NH_3)_5Cl]^{2+}$; 535 nm

CCC378

4. The wave length of light absorbed is highest in :

(A) $[Co(NH_3)_5Cl]^{2\oplus}$ (B) $[Co(NH_3)_5H_2O]^{3\oplus}$ (C) $[Co(NH_3)_6]^{3\oplus}$ (D) $[Co(en)_3]^{3\oplus}$

CCC305

5. If a transition – metal compound absorbs violet – indigo radiation in the visible region. Its colour would be

(A) Green (B) Yellow (C) Orange (D) Blue

CCC306

Bonding in Metal Carbonyls

6. $[Co_2(CO)_8]$ displays :

(A) No Co-Co bond, four terminal CO and four bridging CO

(B) One Co-Co bond, six terminal CO and two bridging CO

(C) No Co-Co bond, six terminal CO and two bridging CO

(D) One Co-Co bond, four terminal CO and four bridging CO

CCC379

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Synergic Bonding, IRVF and Stretching frequency, Organometallic compounds

7. The V–C distance in $V(CO)_6$ and $[V(CO)_6]^{\ominus}$ are respectively (in pm)-

(A) 200, 200 (B) 193, 200 (C) 200, 193 (D) 193, 193

CCC134

8. Ziegler-Natta catalyst is ?

(A) Pt / PtO (B) $Al(C_2H_5)_3 + TiCl_4$

(C) $K[PtCl_3(\eta^2 - C_2H_4)]$ (D) $K[PtCl_3(\eta^2 - C_2H_4)]$

CCC185

9. Among the following, metal carbonyls, the C–O bond is strongest :

(A) $[Mn(CO)_6]^+$ (B) $[Cr(CO)_6]$ (C) $[V(CO)_6]^-$ (D) $[Ti(CO)_6]^{2-}$

CCC380

10. What is the **CORRECT** order of C—O bond length?

(A) $(C—O)_{Fe(CO)_5} > (C—O)_{CO} > (C—O)_{CO^+}$ (B) $(C—O)_{Fe(CO)_5} > (C—O)_{CO^+} > (C—O)_{CO}$

(C) $(C—O)_{CO^+} > (C—O)_{CO} > (C—O)_{Fe(CO)_5}$ (D) $(C—O)_{CO^+} > (C—O)_{Fe(CO)_5} > (C—O)_{CO}$

CCC381

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	C	A	A	B	B	C	B	A	A

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